

EXERCISE - O

SINGLE CORRECT TYPE QUESTIONS

Introduction of Metallurgy, types of ores

1. Which of the following is not an ore of aluminium?
 (A) Bauxite (B) Corundum (C) Limonite (D) Kaolinite CMT102
2. Which of the following set of metals are mostly found as sulphide ores :
 (A) Zn, Cu, Mg (B) Zn, Cu, Pb (C) Fe, Al, Sn (D) Cu, Ag, Au CMT103
3. Which mineral has been named incorrectly?
 (A) Bauxite : $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
 (B) Corundum : Al_2O_3
 (C) Cryolite : $3\text{NaF} \cdot \text{AlF}_3$
 (D) Fluorspar : $\text{Be}_3\text{Al}_2\text{Si}_6\text{O}_{18}$ CMT104
4. Select correct statement
 (a) Aluminium is third most abundant element in earth crust.
 (b) Aluminium is major component of many igneous minerals including mica and clays.
 (c) Many gem stones are impure forms of Al_2O_3
 (d) Iron is the second most abundant metal in the earth crust
 (A) Only a, b and c (B) Only a & c (C) a, b, c and d (D) Only a, b & d CMT107
5. The process of the isolation of a metal by dissolving the ore in aqueous solution of suitable chemical reagent followed by precipitation of the metal by a more electropositive metal is called:
 (A) hydrometallurgy (B) electrometallurgy
 (C) zone refining (D) electrorefining CMT012

Metallurgical Processes

6. Gravity separation process may be used for the concentration of :
 (A) Copper pyrites (B) Bauxite (C) Haematite (D) Argentite CMT112
7. Which of the following ores is concentrated by magnetic separation method ?
 (A) Bauxite (B) Chromite (C) Argentite (D) Dolomite CMT114
8. Which one of the following reactions represents a calcination reaction?
 (A) $\text{HgS} + \text{O}_2 \rightarrow \text{Hg} + \text{SO}_2$ (B) $2\text{CuFeS}_2 + 4\text{O}_2 \rightarrow \text{Cu}_2\text{S} + 2\text{FeO} + 3\text{SO}_2$
 (C) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2 \rightarrow \text{CuO} + \text{CO}_2 + \text{H}_2\text{O}$ (D) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} + 2\text{NaOH} \rightarrow 2\text{NaAlO}_2 + 3\text{H}_2\text{O}$ CMT119

9. Select incorrect statement regarding calcination?

- (A) In calcination usually, the hydroxide ore and carbonate ore are converted to oxide.
- (B) It is an endothermic process.
- (C) It is carried out in absence of air.
- (D) After calcination surface area of ore decreases.

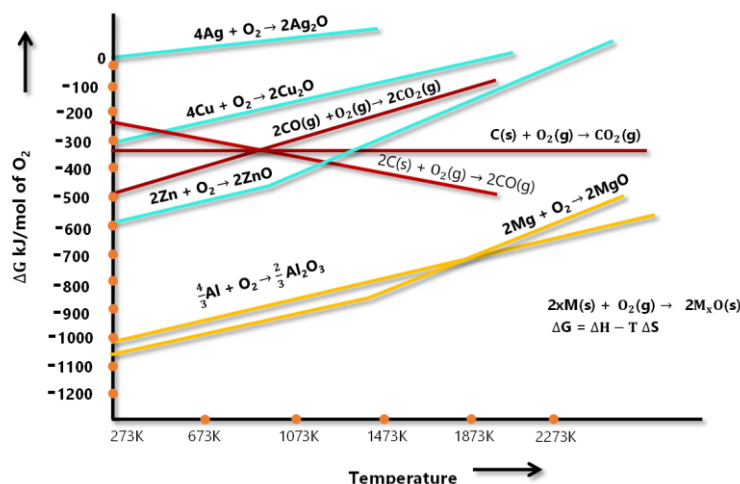
CMT121

10. Incorrect statement regarding roasting is :

- (A) Usually the sulphide ore is converted to oxide before reduction because oxides ore easier to reduce.
- (B) It is an endothermic process.
- (C) It is carried out in reverberatory furnace.
- (D) Slag formation may take place during roasting.

CMT123

11. Considering Ellingham diagram, which of the following metals can be used to reduce alumina ?



- (A) Fe
- (B) Cu
- (C) Mg
- (D) Zn

CMT125

12. Which of the following metals cannot be extracted by the carbon reduction process?

- (A) Zn
- (B) Fe
- (C) Al
- (D) Sn

CMT127

13. Carbon cannot be used in the reduction of Al_2O_3 because :

- (A) It is non-metal
- (B) The heat of formation of CO_2 is more than that of Al_2O_3
- (C) Pure carbon is not easily available
- (D) The heat of formation of Al_2O_3 is too high

CMT129

14. Ellingham diagram represents:

- (A) Change of ΔG° with temperature.
- (B) Change of ΔH° with temperature.
- (C) Change of ΔG° with pressure.
- (D) Change of $(\Delta G^\circ - T\Delta S^\circ)$ with temperature.

CMT130

15. Which of the following metals is obtained by the self reduction process?

- (A) Copper
- (B) Tin
- (C) Gold
- (D) Magnesium

CMT132

16. Identify the metal M whose extraction is based on the following reactions :
 $MS + 2O_2 \rightarrow MSO_4$
 $2MS + 3O_2 \rightarrow 2MO + 2SO_2$
 $MS + 2MO \rightarrow 3M + SO_2$
 $MS + MSO_4 \rightarrow 2M + 2SO_2$
 (A) Zinc (B) Aluminium (C) Lead (D) Silver **CMT134**
17. In which of the following pair of metals, both are commercially extracted from their respective ores by self reduction method ?
 (A) Zn, Cu (B) Pb, Cu (C) Sn, Zn (D) Al, Ag **CMT136**
18. Poling process :
 (A) Reduces SnO_2 to Sn.
 (B) Involves the liberation of reducing gases like hydrocarbons.
 (C) Uses poles of freshly cut green wood.
 (D) All of the above are correct. **CMT138**
19. The method of zone refining of metals is used for the purification of :
 (A) Zn (B) Al (C) Cu (D) None **CMT140**
20. In the electrolytic refining of lead, Sb, Cu, Ag and Au are found :
 (A) On anode (B) In electrolyte solution
 (C) In anode mud (D) In cathode mud **CMT144**
21. Van Arkel method of purification of metals involves converting the metal to:
 (A) Volatile stable compound. (B) Volatile unstable compound.
 (C) Non-volatile stable compound. (D) None of these. **CMT148**
22. The iron obtained from the blast furnace is called :
 (A) Pig iron (B) Cast iron (C) Wrought iron (D) Steel **CMT167**
23. Process used for extraction of Ag from its native ore is :
 (A) Self reduction (B) Carbon reduction
 (C) Leaching and Hydrometallurgy (D) Electrolytic reduction **CMT181**
24. Which of the following metals cannot be extracted by the carbon reduction process ?
 (A) Zn (B) Fe (C) Al (D) Sn **CMT002**
25. In Ellingham diagram which of the following reaction gives line almost parallel to x-axis ?
 (A) $CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$ (B) $C(s) + O_2(g) \rightarrow CO_2(g)$
 (C) $C(s) + \frac{1}{2} O_2(g) \rightarrow CO(g)$ (D) $2Al(s) + \frac{1}{2} O_2(g) \rightarrow Al_2O_3(s)$ **CMT005**

26. An ore of metal 'M' $\xrightarrow{\text{Roasting}}$ Metal oxide
 (After crushing and grinding)
 $\downarrow$ Carbon Reduction
 'M'
- 'M' may be :-
 (A) Al (B) Zn (C) Hg (D) All of these CMT007
27. Which of the following cannot be reduced by carbon ?
 (A) PbO (B) SnO₂ (C) ZnO (D) MgO CMT008
28. The following equation represents a method of purification of nickel by :-

$$\text{Ni (impure)} + 4\text{CO} \xrightarrow{320\text{K}} \text{Ni(CO)}_4 \xrightarrow{420\text{K}} \text{Ni (pure)} + 4\text{CO}$$
 (A) Cupellation (B) Mond's process
 (C) Van Arkel method (D) Zone refining CMT010
29. Refractory materials are generally used in furnaces because
 (A) they are chemically inert (B) they can withstand high temperature
 (C) they do not contain impurities (D) they decrease melting point of ore CMT019
30. Which of the following statements is correct regarding the slag formation during the extraction of a metal like copper or iron.
 (A) The slag is lighter and has lower melting temperature than the metal
 (B) The slag is heavier and has lower melting temperature than the metal
 (C) The slag is lighter and has higher melting temperature than the metal
 (D) The slag is heavier and has higher melting temperature than the metal CMT020

Extraction of metals

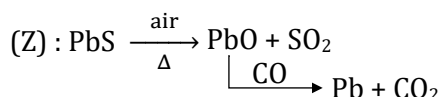
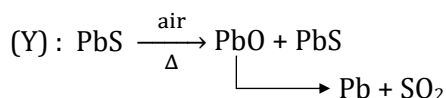
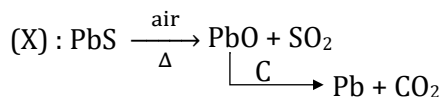
31. Refining of Gold is done by :
 (A) Parke's process (B) Parting process (C) Polling process (D) Kroll's process CMT096
32. Extraction of gold and silver involves leaching with $\text{CN}^\ominus$ ion. Silver is later recovered by :
 (A) Liquation (B) Distillation (C) Zone refining (D) Displacement with Zn CMT118
33. To purify the 'blister copper' it is melted in a furnace and then stirred with the poles of freshly cut green wood. The purpose of this is :
 (A) To remove the gases dissolved in blister copper.
 (B) To bring the impurities to the surface of the molten liquid.
 (C) To increase the carbon percentage.
 (D) To reduce the metallic oxide to metal by hydrocarbons liberated by green wood CMT142

34. In electrolytic refining of lead, electrolyte consists of :
(A) H_2SiF_6 only (B) PbSiF_6 only (C) H_2SiF_6 only (D) H_2SiF_6 and PbSiF_6
CMT146
35. In electrolysis of Al_2O_3 by Hall-Heroult process :
(A) Cryolite $\text{Na}_3[\text{AlF}_6]$ lowers the melting point of Al_2O_3 and increases its electrical conductivity.
(B) Al is obtained at cathode and probably CO_2 at anode.
(C) Oxygen liberated at anode reacts with carbon anode liberating CO and CO_2 .
(D) All of the above are correct.
CMT152
36. In the electrolytic reduction of fused mixture of Al_2O_3 and Na_3AlF_6 gas liberated at graphite anode is
(A) F_2 (B) OF_2 (C) O_2 (D) O_3
CMT154
37. Aluminium metal is purified by :
(A) Hoopes's electrolytic reduction process (B) Dow's process
(C) Serpeck's process (D) Baeyer's process
CMT156
38. Roasting of ZnS is done at high temperature and in controlled air current to -
(A) Prevent formation of ZnO (B) Prevent formation of ZnSO_4
(C) Prevent formation of Zn (D) None of these
CMT158
39. Blister copper is :
(A) Impure copper.
(B) Obtained in self-reduction process during bessemerisation.
(C) Both (A) and (B) are correct
(D) None of the above
CMT160
40. Which of the following statement is incorrect about the extractive metallurgy of copper?
(A) Matte chiefly consists of iron sulphide and some ferrous oxide
(B) The impurity of iron oxide is removed as fusible slag, FeSiO_3 during roasting
(C) The copper pyrite is concentrated by froth floatation process
(D) Copper is obtained by self-reduction in Bessemer converter
CMT162
41. Identify M and N in the following reaction :
Copper glance $\xrightarrow{\text{Controlled heating in presence of air}}$ M + SO_2
 $\text{M} \xrightarrow{\text{N}} \text{Cu} + \text{SO}_2 (\uparrow)$
(A) $\text{M} = \text{Cu}_2\text{O} + \text{Cu}_2\text{S}$, N = only heating
(B) $\text{M} = \text{Cu}_2\text{O}$, N = Carbon reduction
(C) $\text{M} = \text{Cu}_2\text{O}$, N = Electrolytic reduction
(D) None of these
CMT163

42. If CaO (flux) is added in excess during smelting of cassiterite, then :
 (A) Calcium stannite will form (B) Ferrous silicate will not form
 (C) Calcium silicate will not form (D) None of these

CMT171

43. Main source of lead is galena. It is converted to Pb by :



Self - reduction process is :

- (A) X (B) Y (C) Z (D) none

CMT173

44. When impurity content is high then lead is extracted from galena by?

- (A) Self reduction (B) Carbon reduction
 (C) Electrolytic reduction (D) Hydrometallurgy

CMT175

45. Process used for extraction of Mg from molten MgCl_2 is :

- (A) Self reduction (B) Carbon reduction
 (C) Electrolytic reduction (D) Hydrometallurgy

CMT177

46. In Dow's process, Sea water is treated with:

- (A) First by Coke then NaOH (B) First by NaCl then Coke
 (C) First by Ca(OH)_2 then HCl (D) First by SiO_2 then Coke

CMT179

47. Lead present in silver is removed by :

- (A) Cupellation process (B) Parke's process
 (C) Mond's process (D) Van Arkel process

CMT183

48. Silver is removed from impure gold by:

- (A) Cupellation (B) Heating with borax
 (C) Heating with conc. H_2SO_4 (D) Heating with Coke

CMT185

49. Blister copper is :

- (A) Impure copper.
 (B) Obtained in self reduction process during bessemerization.
 (C) Both (A) and (B) are correct.
 (D) None is correct.

CMT001

50. During the electrolytic reduction of aluminium, the carbon anodes are replaced from time to time because:
(A) the carbon anodes get decayed
(B) the carbon prevents atmospheric oxygen from coming in contact with aluminium
(C) oxygen liberated at the carbon anodes reacts with anodes to form CO and CO₂
(D) carbon converts Al₂O₃ to Al
CMT003
51. Silver can be separated from lead by :
(A) fractional crystallisation (B) liquation
(C) cupellation (D) addition of zinc (Parke's method)
CMT004
52. Which of the following is correct regarding extraction of Zn.
(A) Reduction of ZnO by CO is favourable at low temp.
(B) Zn is obtained as molten layer with slag.
(C) Reduction of ZnO by CO is reversible and high temperature is required.
(D) In reduction process CO₂ is the only gaseous product.
CMT006
53. In the Down's process of obtaining sodium, CaCl₂ is added -
(A) To get high yield of sodium (B) To improve the quality of sodium obtained
(C) To lower the operating temperature (D) To act as a reducing agent
CMT009
54. $\text{Ag}_2\text{S} + \text{NaCN} + \text{Zn} \longrightarrow \text{Ag}$
This method of extraction of Ag by complex formation and then its displacement is called:
(A) Parke's method (B) McArthur-Forest method
(C) Serpeck method (D) Hall's method
CMT011
55. Blister copper is refined by stirring molten impure metal with green logs of wood because such a wood liberates hydrocarbon gases (like CH₄). This process X is called _____ and the metal contains impurities of Y is _____.
(A) X = cupellation, Y = CuO₂ (B) X = poling, Y = Cu₂O
(C) X = poling, Y = CuO (D) X = cupellation, Y = CuO
CMT013
56. A piece of steel is heated until redness and then plunged into cold water or oil. This treatment of steel makes it
(A) soft and malleable (B) hard but not brittle
(C) more brittle (D) hard and brittle
CMT014
57. Modern method of steel manufacturing is
(A) open hearth process (B) L.D. Process
(C) Bessemerisation (D) Cupellation
CMT015
58. During electrolytic reduction of alumina, two auxiliary electrolytes X and Y are added to increase the electrical conductance and lower the temperature of melt in order to making fused mixture very conducting. X and Y are
(A) cryolite and fluorspar (B) cryolite and alum
(C) alum and fluorspar (D) fluorspar and bauxite
CMT016

59. Railway wagon axles are made by heating iron rods embedded in charcoal powder. This process is known as:
 (A) Sherardising (B) Annealing (C) Tempering (D) Case hardening

CMT017

60. Which of the following statement is correct regarding Cu-extraction
 (A) In the smelting step carbon reduction takes places
 (B) During partial roasting Cu_2S remains almost unaffected
 (C) In Bessemer converter, only self-reduction occur, not slag formation
 (D) Blister forms in the blister Cu is due to dissolved CO_2

CMT018

MULTIPLE CORRECT TYPE QUESTIONS

Metallurgical processes

61. Roasting of copper pyrites is done :
 (A) To remove moisture
 (B) To oxidise free Sulphur and Antimony
 (C) To convert pyrites completely into Cu_2O and FeO
 (D) To remove volatile organic impurities

CMT168

MATRIX MATCH TYPE QUESTION

Introduction of Metallurgy types of ores

62. Match Column-I with Column-II and select the correct answer using the codes given below :

Column-I (Metals)

- (1) Tin
 (2) Zinc
 (3) Copper
 (4) Lead

Codes :

	(1)	(2)	(3)	(4)
(A)	p	q	r	s
(B)	q	p	s	r
(C)	s	r	q	p
(D)	q	p	r	s

Column-II (Ores)

- (p) Zincite
 (q) Cassiterite
 (r) Cerussite
 (s) Cuprite

CMT109

Metallurgical processes

63. Match the method of concentration of the ore in column I with the ore in column II and select the correct alternate :

(I)

- (1) Magnetic separation
 (2) Froth floatation
 (3) Gravity separation

	1	2	3
(A)	A	B	C
(B)	B	A	C
(C)	C	A	B
(D)	B	C	A

(II)

- (A) Cu_2S
 (B) Fe_2O_3
 (C) SnO_2

CMT116

64. Match **Column-I** with **Column-II****Column-I (Metals)**

- (A) Iron & copper
 (B) Zirconium & Titanium
 (C) Lead & Tin
 (D) Copper & Tin

Column-II (Method used for refining)

- (P) Poling
 (Q) Bessemerisation
 (R) Van-Arkel
 (S) Liquation

CMT187

Miscellaneous/Mixed65. **Metal**

- (a) Cu
 (b) Ga
 (c) Ni
 (d) Zr
 (A) a – iii, b – iv, c – ii, d – i
 (C) a – i, b – ii, c – iii, d – iv

Purification Method

- (i) Van arkel process
 (ii) Mond process
 (iii) Electrolytic refining
 (iv) Zone refining method
 (B) a – ii, b – iii, c – iv, d – i
 (D) a – iii, b – iv, c – i, d – ii

CMT150

66. Match the following choosing one item from column X and the appropriate item from column Y.

Column -X

- (A) Zinc from ZnCO_3
 (B) Lead from PbS
 (C) Cu from CuFeS_2
 (D) Tin from cassiterite

Column-Y

- (P) Calcination
 (Q) Removal of iron
 (R) Froth floatation process
 (S) Poling

CMT188

67. Match column - I and column - II and select the correct answer using the codes given below the lists:

Column - I

- (A) Cyanide process
 (B) Floatation process
 (C) Electrolytic reduction
 (D) Zone refining

Column - II

- (P) Ultrapure Ge
 (Q) Dressing of HgS
 (R) Extraction of Al
 (S) Extraction of Au

CMT189

68. Match the items of Column I with items of Column II and assign the correct code :

Column I

- (P) Blistered Cu
 (Q) Blast furnace
 (R) Reverberatory furnace
 (S) Hall-Heroult process

Column II

- (1) Aluminium
 (2) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$
 (3) Iron
 (4) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
 (5) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$

Code :

- (A) P $\rightarrow$ (2) ; Q $\rightarrow$ (3) ; R $\rightarrow$ (4) ; S $\rightarrow$ (1)
 (B) P $\rightarrow$ (1) ; Q $\rightarrow$ (2) ; R $\rightarrow$ (3) ; S $\rightarrow$ (5)
 (C) P $\rightarrow$ (5) ; Q $\rightarrow$ (4) ; R $\rightarrow$ (3) ; S $\rightarrow$ (2)
 (D) P $\rightarrow$ (4) ; Q $\rightarrow$ (5) ; R $\rightarrow$ (3) ; S $\rightarrow$ (2)

CMT190

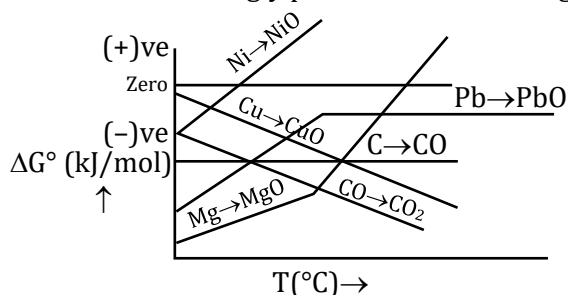
EXERCISE - S

Introduction of Metallurgy, types of ores

- In extraction of metal how many of the following ores calcination process is involved.
 (i) Dolomite (ii) Malachite (iii) Calcite (iv) Copper pyrites
 (v) Sylvine (vi) Cryolite (vii) Siderite (viii) Iron pyrite
 (ix) Argentite CMT021
- How many of the following minerals containing Mg.
 Magnetite, Carnallite, Epsom salt, Siderite CMT028
- Find out the number of minerals given below contain iron as Fe(II).
 Haematite, Magnetite, Limonite, Siderite, Chromite, Wolframite CMT029
- How many of the following ores of silver ?
 Hornsilver, Cerrusite, Chalcopyrite, Galena, Anglesite, Argentite CMT030
- Amongst the following, total number of sulphide ores are
 Calamine, Sphalertie, Copper pyrites, Copper glance, Iron pyrites, Bauxite CMT191

Metallurgical processes

- How many of the following metallurgies given below involve leaching ?
 $\text{Al}_2\text{O}_3 \longrightarrow \text{Al}$; $\text{Ag}_2\text{S} \longrightarrow \text{Ag}$; $\text{Au(ore)} \longrightarrow \text{Au}$; $\text{CuFeS}_2 \longrightarrow \text{Cu}$; $\text{PbS} \longrightarrow \text{Pb}$
 $\text{MgCl}_2 \longrightarrow \text{Mg}$; $\text{FeCO}_3 \longrightarrow \text{Fe}$; Low grade copper ore $\longrightarrow \text{Cu}$; $\text{HgS} \longrightarrow \text{Hg}$ CMT022
- How many of these metals can be purified by vapour phase refining Ni, Ti, Zr, Ge, In, Fe, Cu, Au. CMT025
- Find the number of ore which are concentrated by magnetic separation method.
 Cassiterite, Copper Glance, Chromite, Cinnabar CMT026
- Find the number of curves which are wrongly presented in the Ellingham diagram.



- Find the number of metals which are commercially extracted by carbon reduction method
 Pb, Fe, Zn, Mg, Al, Na, Au, Ag CMT027
- Amongst the following ores, the total number of oxide ores are
 Siderite, Magnetite, Hematite, Malachite, Zincite, Cuprite CMT192
- Amongst the following ores, the total number of oxide ores are
 Siderite, Magnetite, Hematite, Malachite, Zincite, Cuprite CMT193

Miscellaneous/Mixed

- Find the total number of process which involve formation of Cl_2 as one of the product/by product
 • Down's cell • Dow's sea- water process • Deacons process
 • Hall Heroult cell • Castner-Kellner cell • Bayer's process
 • Ostwald process CMT023
- The number of following pairs is correctly matched
 (i) Van Arkel method – Zirconium (ii) Mond Process – Titanium
 (iii) Froth Floatation Method – Cerussite (iv) Distillation method – Zinc
 (v) Poling Process – Copper (vi) Mac Aurther process – Gold CMT024

EXERCISE - JEE (Main) PYQ

1. The substance used as froth stabilisers in froth-floatation process is :
[JEE (Main) 2012]
(1) Copper sulphate (2) Aniline
(3) Sodium cyanide (4) Potassium ethyl xanthate
CMT031
2. Which of the oxide groups among the following cannot be reduced by carbon:-
[JEE (Main) 2012]
(1) Fe_3O_4 , ZnO (2) PbO , Fe_3O_4 (3) Cu_2O , SnO_2 (4) CaO , K_2O
CMT032
3. In Goldschmidt aluminothermite process which of the following reducing agents is used :
[JEE (Main) 2013]
(1) Calcium (2) Coke (3) Sodium (4) Al-powder
CMT033
4. Calcination is the process in which :
[JEE (Main) 2013]
(1) Ore is heated strongly below its melting point in the presence of excess of air and is used for the conversion of carbonates and hydrated oxide ores to their respective oxides.
(2) Ore is heated strongly below its melting point in the absence or limited supply of air and is used for conversion of sulphide ores to their respective oxides
(3) Ore is heated strongly below its melting point either in the limited supply or absence of air and is used to convert carbonates and hydrated oxide ores to their respective oxides
(4) Ore is heated strongly above its melting point in the limited supply of air to convert sulphide ores to their respective oxides.
CMT034
5. The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is :
[JEE (Main) 2014]
(1) Cu (2) Cr (3) Ag (4) Ca
CMT035
6. The form of iron obtained from blast furnace is :
[JEE (Main) 2014]
(1) Steel (2) Wrought Iron (3) Cast Iron (4) Pig iron
CMT036
7. In the context of the Hall-Heroult process for the extraction of Al, which of the following statements is false ?
[JEE (Main) 2015]
(1) Al^{3+} is reduced at the cathode to form Al
(2) Na_3AlF_6 serves as the electrolyte
(3) CO and CO_2 are produced in this process
(4) Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity
CMT037

8. Which one of the following ores is best concentrated by froth floatation method?

[JEE (Main) 2016]

- (1) Malachite (2) Magnetite (3) Siderite (4) Galena

CMT038

9. What will occur if a block of copper metal is dropped into a beaker containing a solution of 1M ZnSO_4 ?

[JEE (Main) 2016]

- (1) The copper metal will dissolve and zinc metal will be deposited
(2) No reaction will occur
(3) The copper metal will dissolve with evolution of oxygen gas
(4) The copper metal will dissolve with evolution of hydrogen gas

CMT039

10. In the leaching method, bauxite ore is digested with a concentrated solution of NaOH that produces 'X'. When CO_2 gas is passed through the aqueous solution of 'X', a hydrated compound 'Y' is precipitated. 'X' and 'Y' respectively are :-

[JEE (Main) 2018]

- (1) $\text{Na}[\text{Al}(\text{OH})_4]$ and $\text{Al}_2\text{O}_3 \cdot x \text{H}_2\text{O}$ (2) $\text{Al}(\text{OH})_3$ and $\text{Al}_2\text{O}_3 \cdot x \text{H}_2\text{O}$
(3) $\text{Na}[\text{Al}(\text{OH})_4]$ and $\text{Al}_2(\text{CO}_3)_3 \cdot x \text{H}_2\text{O}$ (4) Na AlO_2 and $\text{Al}_2(\text{CO}_3)_3 \cdot x \text{H}_2\text{O}$

CMT040

11. When metal 'M' is treated with NaOH, a white gelatinous precipitate 'X' is obtained, which is soluble in excess of NaOH. Compound 'X' when heated strongly gives an oxide which is used in chromatography as an adsorbent. The metal 'M' is

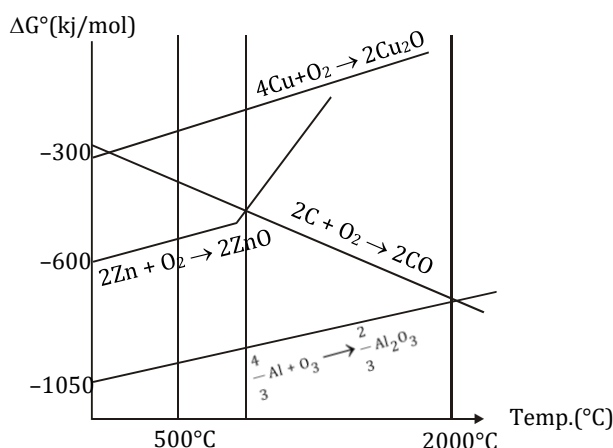
[JEE (Main) 2018]

- (1) Ca (2) Al (3) Fe (4) Zn

CMT041

12. The correct statement regarding the given Ellingham diagram is:

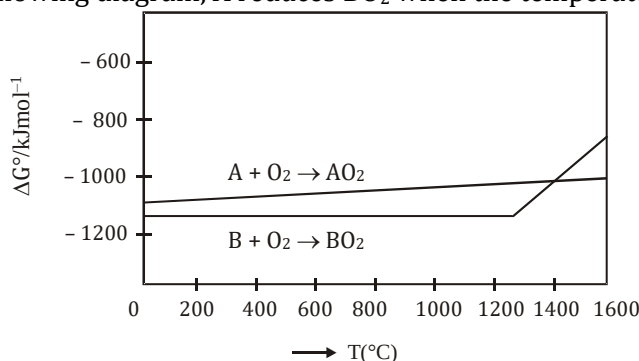
[JEE (Main) 2019]



- (1) At 800°C, Cu can be used for the extraction of Zn from ZnO
(2) At 500°C, coke can be used for the extraction of Zn from ZnO
(3) Coke cannot be used for the extraction of Cu from Cu_2O .
(4) At 1400°C, Al can be used for the extraction of Zn from ZnO

CMT042

13. According to the following diagram, A reduces BO_2 when the temperature is : [JEE (Main) 2020]



- (1) $< 1400^\circ\text{C}$ (2) $> 1400^\circ\text{C}$ (3) $< 1200^\circ\text{C}$ (4) $> 1200^\circ\text{C}$ but $< 1400^\circ\text{C}$

CMT043

14. The processes of calcination and roasting in metallurgical industries, respectively, can lead to :- [JEE (Main) 2020]
- (1) Global warming and acid rain
 - (2) Photochemical smog and ozone layer depletion
 - (3) Global warming and photochemical smog
 - (4) Photochemical smog and global warming

CMT044

15. Which of the following ore is concentrated using group 1 cyanide salt ? [JEE (Main) 2021]
- (1) Sphalerite (2) Calamine (3) Siderite (4) Malachite

CMT045

16. Match List-I with List-II : [JEE (Main) 2021]

List-I**Industrial process**

- (a) Haber's process
- (b) Ostwald's process
- (c) Contact process
- (d) Hall-Heroult process

List-II**Application**

- (i) HNO_3 synthesis
- (ii) Aluminium extraction
- (iii) NH_3 synthesis
- (iv) H_2SO_4 synthesis

Choose the correct answer from the options given below :

- (1) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
- (2) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)
- (3) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
- (4) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

CMT046

17. The metal that can be purified economically by fractional distillation method is: [JEE (Main) 2021]

- (1) Fe (2) Zn (3) Cu (4) Ni

CMT047

18. Given below are two statements.
Statement I: The choice of reducing agents for metals extraction can be made by using Ellingham diagram, a plot of ΔG vs temperature.
Statement II: The value of ΔS increases from left to right in Ellingham diagram.
 In the light of the above statements, choose the **most appropriate** answer from the options given below: [JEE (Main) 2021]
- (1) Both **Statement I** and **Statement II** are true
 - (2) **Statement I** is false but **Statement II** is true
 - (3) Both **Statement I** and **Statement II** are false
 - (4) **Statement I** is true but **Statement II** is false

CMT048

19. Match List - I with List - II

[JEE (Main) 2022]

	List - I		List - II
(A)	Sphalerite	(I)	FeCO_3
(B)	Calamine	(II)	PbS
(C)	Galena	(III)	ZnCO_3
(D)	Siderite	(IV)	ZnS

Choose the most appropriate answer from the options given below:

- (1) (A) - (IV), (B) - (III), (C) - (II), (D) - (I) (2) (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
 (3) (A) - (II), (B) - (III), (C) - (I), (D) - (IV) (4) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

CMT049

20. The compound(s) that is(are) removed as slag during the extraction of copper is :

[JEE (Main) 2022]

- (1) CaO (2) FeO (3) Al_2O_3 (4) ZnO
 (5) NiO

Choose the correct answer from the options given below :

- (1) (3) (4) Only (2) (1), (2), (5) Only
 (3) (1), (2) Only (4) (2) Only

CMT050

21. Match List I with List II.

[JEE (Main) 2023]

List - I		List - II	
A.	Reverberatory furnace	I.	Pig Iron
B.	Electrolytic cell	II.	Aluminium
C.	Blast furnace	III.	Silicon
D.	Zone Refining furnace	IV.	Copper

- (1) A - IV, B - II, C - I, D - III (2) A - I, B - IV, C - II, D - III
 (3) A - I, B - III, C - II, D - IV (4) A - III, B - IV, C - I, D - II

CMT051

EXERCISE - JEE (Advanced) PYQ

1. In extractive metallurgy of zinc partial fusion of ZnO with coke is called ____ and reduction of the ore to the molten metal is called ____ (smelting, calcining, roasting, sintering).
[JEE (Advanced) 1988]
CMT072
2. The methods chiefly used for the extraction of lead and tin from their ores are respectively :
[JEE (Advanced) 2004]
(A) self reduction and carbon reduction (B) self reduction and electrolytic reduction
(C) carbon reduction and self reduction (D) cyanide process and carbon reduction
CMT073
3. Which ore contains both iron and copper?
[JEE (Advanced) 2004]
(A) Cuprite (B) Chalcocite (C) Chalcopyrite (D) Malachite
CMT074
4. Match the extraction processes listed in column I with metals listed in column II.
[JEE (Advanced) 2006]
- | Column I | Column II |
|---|---------------------------------------|
| (A) Self reduction | (P) Lead |
| (B) Carbon reduction | (Q) Silver |
| (C) Complex formation and displacement by metal | (R) Copper |
| (D) Hall Heroult process | (S) Aluminium |
| (A) ((A)-P,Q ; (B)-R ; (C)-Q ; (D)-S) | (B) ((A)-P,R ; (B)-P ; (C)-Q ; (D)-S) |
| (C) ((A)-R,S ; (B)-P ; (C)-Q ; (D)-P) | (D) ((A)-P,Q ; (B)-R ; (C)-Q ; (D)-P) |
- CMT075
5. Extraction for zinc from zinc blende is achieved by :
[JEE (Advanced) 2007]
(A) electrolytic reduction
(B) roasting followed by reduction with carbon
(C) roasting followed by reduction with another metal
(D) roasting followed by self-reduction
CMT076
6. Match the conversions in **Column I** with the type(s) of reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4×4 matrix given in the ORS.
[JEE (Advanced) 2008]
- | Column I | Column II |
|---|---------------------------------------|
| (A) $\text{PbS} \rightarrow \text{PbO}$ | (P) Roasting |
| (B) $\text{CaCO}_3 \rightarrow \text{CaO}$ | (Q) Calcination |
| (C) $\text{ZnS} \rightarrow \text{Zn}$ | (R) Carbon reduction |
| (D) $\text{Cu}_2\text{S} \rightarrow \text{Cu}$ | (S) Self reduction |
| (A) (A)-P,S ; (B)-S ; (C)-Q,R ; (D)-P | (B) (A)-R ; (B)-Q ; (C)-P ; (D)-P,R |
| (C) (A)-P ; (B)-S ; (C)-R ; (D)-P | (D) (A)-P ; (B)-Q ; (C)-P,R ; (D)-P,S |
- CMT077
7. Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of :-
[JEE (Advanced) 2008]
(A) nitrogen (B) oxygen (C) carbon dioxide (D) argon
CMT078

Paragraph for questions 8 to 10

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcantite ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), atacamite ($\text{Cu}_2\text{Cl}(\text{OH})_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($\text{Cu}_2(\text{OH})_2\text{CO}_3$). However, 80% of the world copper production comes from the ore chalcopyrite (CuFeS_2). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction.
[JEE (Advanced) 2010]

8. Partial roasting of chalcopryrite produces :-
 (A) Cu_2S and FeO (B) Cu_2O and FeO (C) CuS and Fe_2O_3 (D) Cu_2O and Fe_2O_3 CMT079
9. Iron is removed from chalcopryrite as :- [JEE (Advanced) 2010]
 (A) FeO (B) FeS (C) Fe_2O_3 (D) FeSiO_3 CMT080
10. In self-reduction, the reducing species is :- [JEE (Advanced) 2010]
 (A) S (B) O^{2-} (C) S^{2-} (D) SO_2 CMT081
11. Extraction of metal from the ore cassiterite involves [JEE (Advanced) 2011]
 (A) carbon reduction of an oxide ore (B) self-reduction of a sulphide ore
 (C) removal of copper impurity (D) removal of iron impurity CMT082
12. Oxidation states of the metal in the minerals haematite and magnetite, respectively, are [JEE (Advanced) 2011]
 (A) II, III in haematite and III in magnetite (B) II, III in haematite and II in magnetite
 (C) II in haematite and II, III in magnetite (D) III in haematite and II, III in magnetite CMT083
13. In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are : [JEE (Advanced) 2012]
 (A) O_2 and CO respectively. (B) O_2 and Zn dust respectively.
 (C) HNO_3 and Zn dust respectively. (D) HNO_3 and CO respectively. CMT084
14. Sulphide ores are common for the metals - [JEE (Advanced) 2013]
 (A) Ag , Cu and Pb (B) Ag , Cu and Sn (C) Ag , Mg and Pb (D) Al , Cu and Pb CMT085
15. The carbon-based reduction method is **NOT** used for the extraction of [JEE (Advanced) 2013]
 (A) tin from SnO_2 (B) Iron from Fe_2O_3
 (C) aluminium from Al_2O_3 (D) magnesium from $\text{MgCO}_3 \cdot \text{CaCO}_3$ CMT086
16. Upon heating with Cu_2S , the reagent(s) that give copper metal is/are [JEE (Advanced) 2014]
 (A) CuFeS_2 (B) CuO (C) Cu_2O (D) CuSO_4 CMT087
17. Copper is purified by electrolytic refining of blister copper. The correct statement(s) about this process is (are) [JEE (Advanced) 2015]
 (A) Impure Cu strip is used as cathode (B) Acidified aqueous CuSO_4 is used as electrolyte
 (C) Pure Cu deposits at cathode (D) Impurities settle as anode-mud CMT088
18. Match the anionic species given in Column-I that are present in the ore(s) given in Column-II [JEE (Advanced) 2015]
Column - I **Column - II**
 (A) Carbonate (P) Siderite
 (B) Sulphide (Q) Malachite
 (C) Hydroxide (R) Bauxite
 (D) Oxide (S) Calamine
 (T) Argentite
 (A) (A) - P, R, S ; (B) - R ; (C) - Q, S ; (D) - Q (B) (A) - P, Q, S ; (B) - T ; (C) - Q, R ; (D) - R
 (C) (A) - P, S ; (B) - P, T ; (C) - Q, S ; (D) - T (D) (A) - Q, R ; (B) - Q, T ; (C) - Q, T ; (D) - Q CMT089

19. Extraction of copper from copper pyrite (CuFeS_2) involves [JEE (Advanced) 2016]
(A) crushing followed by concentration of the ore by froth-flotation
(B) removal of iron as slag
(C) self-reduction step to produce 'blister copper' following evolution of SO_2
(D) refining of 'blister copper' by carbon reduction
CMT090
20. Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of O_2 consumed is Medium.
(Atomic weights in g mol : O = 16, S = 32, Pb = 207) [JEE (Advanced) 2018]
CMT091
21. Calamine, malachite, magnetite and cryolite, respectively are [JEE (Advanced) 2019]
(A) ZnSO_4 , CuCO_3 , Fe_2O_3 , AlF_3
(B) ZnCO_3 , $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$, Fe_3O_4 , Na_3AlF_6
(C) ZnSO_4 , $\text{Cu}(\text{OH})_2$, Fe_3O_4 , Na_3AlF_6
(D) ZnCO_3 , CuCO_3 , Fe_2O_3 , Na_3AlF_6
CMT092
22. The cyanide process of gold extraction involves leaching out gold from its ore with CN^- in the presence of Q in water to form R. Subsequently, R is treated with T to obtain Au and Z. Choose the correct option(s). [JEE (Advanced) 2019]
(A) T is Zn (B) R is $[\text{Au}(\text{CN})_4]^-$ (C) Z is $[\text{Zn}(\text{CN})_4]^{2-}$ (D) Q is O_2
CMT093
23. Which among the following statement(s) is(are) true for the extraction of aluminium from bauxite? [JEE (Advanced) 2020]
(A) Hydrated Al_2O_3 precipitates, when CO_2 is bubbled through a solution of sodium aluminate.
(B) Addition of Na_3AlF_6 lowers the melting point of alumina.
(C) CO_2 is evolved at the anode during electrolysis.
(D) The cathode is a steel vessel with a lining of carbon.
CMT094
24. The correct statement(s) related to the metal extraction processes is(are) [JEE (Advanced) 2021]
(A) A mixture of PbS and PbO undergoes self-reduction to produce Pb and SO_2 .
(B) In the extraction process of copper from copper pyrites, silica is added to produce copper silicate.
(C) Partial oxidation of sulphide ore of copper by roasting, followed by self-reduction produces blister copper.
(D) In cyanide process, zinc powder is utilized to precipitate gold from $\text{Na}[\text{Au}(\text{CN})_2]$
CMT097
25. The electrochemical extraction of aluminum from bauxite ore involves. [JEE (Advanced) 2022]
(A) the reaction of Al_2O_3 with coke (C) at a temperature $> 2500^\circ\text{C}$.
(B) the neutralization of aluminate solution by passing CO_2 gas to precipitate hydrated alumina ($\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$)
(C) the dissolution of Al_2O_3 in hot aqueous NaOH.
(D) the electrolysis of Al_2O_3 mixed with Na_3AlF_6 to give Al and CO_2 .
CMT098
26. The correct option(s) related to the extraction of iron from its ore in the blast furnace operating in the temperature range 900 – 1500 K is(are) [JEE (Advanced) 2022]
(A) Limestone is used to remove silicate impurity.
(B) Pig iron obtained from blast furnace contains about 4% carbon.
(C) Coke (C) converts CO_2 to CO.
(D) Exhaust gases consist of NO_2 and CO.
CMT099

JEE (Advanced) Practice Paper

(This paper is for yourself practice and assessment the discussion of this paper is optional though you can see PDF solutions or video solutions or solutions in hardcopy whichever is provided.)

SECTION-I

- This section contains **TWO** question.
- Each question has FOUR options (A), (B), (C) and (D). ONLY ONE of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:
 Full Marks : +3, if only the bubble corresponding to the correct option is darkened.
 Zero Marks : 0, if none of the bubbles is darkened.
 Negative Marks : -1, in all other cases

1. Metals and their methods of refinement are given
 (i) Ni, Cu, Zr, Ga
 (ii) Electrolysis, van-Arkel process, Zone refining, Mond's process
 Choose the right method for each.
 (A) Ni : Electrolysis, Cu : van-Arkel process, Zr : Zone refining, Ga : Mond's process
 (B) Ni : Mond's process, Cu : Electrolysis, Zr : van-Arkel process, Ga : Zone refining
 (C) Ni : Mond's process, Cu : van-Arkel process, Zr : Zone refining, Ga : Electrolysis
 (D) Ni : Electrolysis, Cu : Zone refining, Zr : van-Arkel process, Ga : Mond's process

CMT052

2. Are electrolytic method of reduction is employed for the preparation of metals that
 (A) Are weakly electropositive (B) Are moderately electropositive
 (C) Are strongly electropositive (D) From acidic oxides

CMT053

SECTION-II

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
 Full Marks : +4, if only (all) the correct option(s) is (are) chosen.
 Partial Marks : +3, if all the four options are correct but ONLY three options are chosen.
 Partial Marks : +2, if three or more options are correct but ONLY two options are chosen, both of which are correct options.
 Partial Marks : +1, if two or more options are correct but ONLY one option is chosen and it is a correct option.
 Zero Marks : 0, if none of the options is chosen (i.e. the question is unanswered).
 Negative Marks : -2 in all other cases.

3. The method of electrolytic refining is suitable in the extraction of :
(A) Aluminium (B) Magnesium (C) Mercury (D) Calcium
CMT054
4. During the process of electrorefining of copper some metals present as impurity settle as anode mud. These are:
(A) Sn (B) Pb (C) Ag (D) Au
CMT055
5. In the manufacturing of metallic sodium by fused salt-electrolysis method (Down's process), small amount of CaCl_2 that added is known as auxiliary electrolyte and is used to
(A) improve the electrical conductance (B) decrease the melting point of electrolyte
(C) stabilise the metallic sodium (D) Increase the temperature of electrolysis
CMT056
6. Hoop's process of purification of aluminium involves formation of layers during electrolysis. It involves
(A) the three layers have same densities but different materials.
(B) the three layers have different densities
(C) the upper layer is of pure aluminium which acts as a cathode
(D) the bottom layer is of impure aluminium which acts as an anode and middle layer consists of cryolite and BaF_2 .
CMT057
7. Metallurgical process of zinc involves roasting of zinc sulphide followed by reduction. Metallic zinc distills over as it is volatile and impurities like Cu, Pb and Fe gets condensed. The crude metal obtained is called spelter, which may be purified by
(A) electrolysis process (B) fractional distillation
(C) polling (D) heating with iodine
CMT058
8. During extraction of copper, it is obtained in the form of molten matte. Which of the following is **not true**?
(A) matte is further treated in Bessemer's converter
(B) molten matte is electrolysed
(C) It is treated with a blast of air and sand
(D) It is dissolved in CuSiF_6 and crystallised.
CMT059
9. The major role of fluorspar (CaF_2) which is added in small quantities in the electrolytic reduction of alumina dissolved in fused cryolite (Na_3AlF_6) is
(A) as a catalyst
(B) to make the fused mixture very conducting
(C) to lower the melting temperature of the mixture
(D) to decrease the rate of oxidation of carbon at the anode.
CMT060

SECTION-III

- This section contains **TWO** question.
- Each question has matching lists. The codes for the lists have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct
- For each question, marks will be awarded in one of the following categories:
 Full Marks : +3, if only the bubble corresponding to the correct option is darkened.
 Zero Marks : 0, if none of the bubbles is darkened.
 Negative Marks : -1, in all other cases.

10. Match column (I) (process) with column (II) (electrolyte)

Column (I) (process)

- (A) Downs cell
 (B) Dow's sea water process
 (C) Hall-Heroult

- (A) (A) Q ; (B) P; (C) R
 (C) (A) R ; (B) P; (C) Q

Column (II) (electrolyte)

- (P) fused MgCl_2
 (Q) fused $(\text{Al}_2\text{O}_3 + \text{Na}_3\text{AlF}_6 + \text{CaF}_2)$
 (R) fused (40% NaCl + 60% CaCl_2)
 (S) $(\text{Al}_2\text{O}_3 + \text{C} + \text{N}_2)$

- (B) (A) P ; (B) P; (C) P
 (D) (A) Q ; (B) P; (C) R

CMT061

11. Match column - I with column - II

Column - I (Property)

- (A) Explosive
 (B) Self-reduction
 (C) Ferrimagnetic material
 (D) Verdigris
 (A) (A) S ; (B) P; (C) Q; (D) R
 (C) (A) R ; (B) S; (C) Q; (D) R

Column - II (Element/compound)

- (P) Cu
 (Q) Fe_3O_4
 (R) $\text{Cu}(\text{CH}_3\text{COO})_2 \cdot \text{Cu}(\text{OH})_2$
 (S) $\text{Pb}(\text{NO}_3)_2$
 (B) (A) Q ; (B) P; (C) S; (D) R
 (D) (A) Q ; (B) P; (C) S; (D) R

CMT062

SECTION-IV

- This section contains **ONE** questions of matching type.
- This section contains **One** table (having 3 columns and 4 rows)
- Based on table, there are **THREE** questions
- Each question has FOUR options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
 Full Marks : +3, if only the bubble corresponding to the correct option is darkened.
 Zero Marks : 0, if none of the bubbles is darkened.
 Negative Marks : -1, in all other cases.

Answer Q.12, Q.13 and Q.14 by appropriately matching the information given in the three columns of the following table.

Column - I Extraction of metal	Column - II Methods for Reduction	Column - III Refining Method
(1) $\text{SnO}_2 \rightarrow \text{Sn}$	(i) Carbon Reduction	(P) Poling
(2) $\text{Al}_2\text{O}_3 \rightarrow \text{Al}$	(ii) Hydro metallurgic Reduction	(Q) Electrolytic Refining
(3) $\text{Cu}_2\text{S} \rightarrow \text{Cu}$	(iii) Electrolytic Reduction	(R) Distillation
(4) $\text{ZnS} \rightarrow \text{Zn}$	(iv) Self-Reduction	(S) Puddling Process

12. Which of the following is **NOT** correctly matched?

- (A) (1), (iv), (S) (B) (3), (iv), (P) (C) (4), (i), (R) (D) (2), (iii), (Q)

CMT063

13. Which of the following match is the **CORRECT** reduction and purification methods for Zn

- (A) (i), (R) (B) (iv), (S) (C) (iv), (P) (D) None of these

CMT064

14. Which of the following set of code shows the **CORRECT** similarity with the extraction processes for Pb?

- (A) (1), (ii), (S) (B) (4), (iii), (P) (C) (2), (iii), (Q) (D) (3), (iv), (Q)

CMT065

SECTION-V

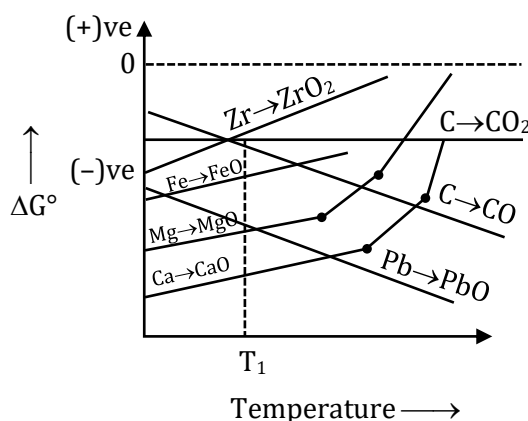
- This section contains **TWO** paragraph.
- Based on each paragraph, there are **TWO/THREE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories:

Full Marks : +3 if only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 in all other cases.

PARAGRAPH FOR QUESTION NO. 15 - 17

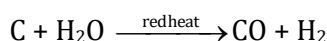
ELLINGHAM DIAGRAM



15. Which of the above curve is wrongly presented -
 (A) $C \rightarrow CO_2$ (B) $Pb \rightarrow PbO$ (C) $Zr \rightarrow ZrO_2$ (D) $Mg \rightarrow MgO$ CMT066
16. Which of the above metal oxide is having minimum thermal decomposition temperature.
 (A) CaO (B) FeO (C) ZrO_2 (D) MgO CMT067
17. Which of the following metal's oxide can be reduced by Fe as reducing agent at temperature (T_1)
 (A) Zr (B) Ca (C) Mg (D) None of these CMT068

PARAGRAPH FOR QUESTION NO. 18 - 19
PURIFICATION METHOD

At high temperature carbon reacts with water to produce a mixture of carbon monoxide, CO and hydrogen, H_2 .



CO is separated from H_2 and then used to separate nickel from cobalt by forming a volatile compound, nickel tetracarbonyl, $Ni(CO)_4$.



18. How many moles of $Ni(CO)_4$ could be obtained from the CO produced by the reaction of 75.0g of carbon ? Assume 100% reaction and 100% recovery in both steps.
 (A) 6.25 (B) 1.563 (C) 3.125 (D) 25.0 CMT069
19. Formation of volatile $Ni(CO)_4$ and its subsequent heating gives pure Ni. Process is called -
 (A) Hall (B) Dow (C) Serpeck (D) Mond CMT070

SECTION-VI

- This section contains **ONE** question.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
 Full Marks : +3, if only the bubble corresponding to the correct answer is darkened.
 Zero Marks : 0, in all other cases.

20. Total number of process which is used for extraction of Aluminium -
 Van arkel process, serpeck's process , baeyer's process , hall's heroult's process CMT071

ANSWER KEY

EXERCISE - O

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	C	A	C	B	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	C	D	A	A	C	B	D	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	A	C	C	B	B	D	B	B	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	B	D	D	D	D	C	A	B	C	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	A	A	B	B	C	C	A	C	C	C
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	D	C	C	B	B	D	B	A	D	B
Que.	61	62	63	64			65	66		
Ans.	ABD	B	C	(A) Q; (B) R; (C) S; (D) P			A	(A) P; (B) R; (C) Q, R, S; (D) Q, S		
Que.	67				68					
Ans.	(A) S ; (B) Q ; (C) R ; (D) P				A					

EXERCISE - S

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	4	2	4	4	3	2	4	3
Que.	11	12	13							
Ans.	4	4	4							

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	4	3	4	4	2	4	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	2	1	1	3	2	4	1	4
Que.	21									
Ans.	1									

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	Sintering, Smelting	A	C	B	B	D	B	A	D	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	AD or ACD	D	B	A	CD	BCD	BCD	B	ABC	6.47
Que.	21	22	23	24	25	26				
Ans.	B	ACD	ABCD	ACD	BCD	ABC				

JEE (Advanced) Practice Paper

Section-I	Q.	1	2						
	A.	B	C						
Section-II	Q.	3	4	5	6	7	8	9	
	A.	ABD	CD	AB	BCD	AB	BD	BC	
Section-III	Q.	10	11						
	A.	C	A						
Section-IV	Q.	12	13	14					
	A.	A	A	D					
Section-V	Q.	15	16	17	18	19			
	A.	B	C	A	B	D			
Section-VI	Q.	20							
	A.	3							